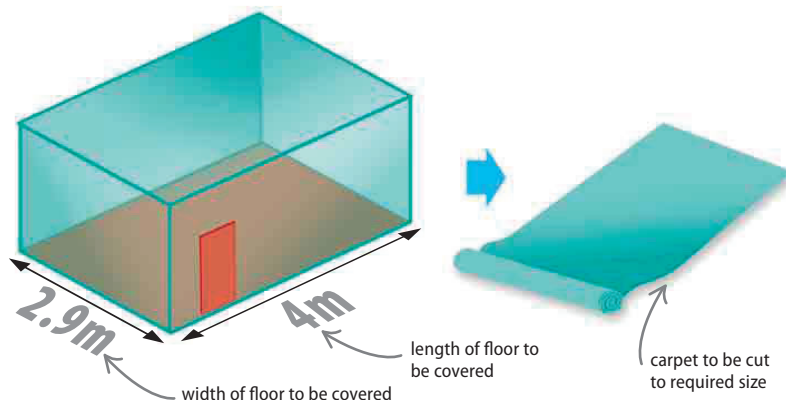


▽ Multiplication using decimals

Decimals appear to complicate the problem, but they can be ignored until the final stage. Here the amount of carpet required to cover a floor needs to be calculated.



LOOKING CLOSER

Checking the answer

Because 2.9 is almost 3, multiplying 3×4 is a good way to check that the calculation to 2.9×4 is correct.

symbol for approximately equal to

$$2.9 \approx 3 \quad \text{and} \quad 3 \times 4 = 12$$

close to real answer of 11.6

$$\text{so } 2.9 \times 4 \approx 12$$

width of floor length

$$2.9 \times 4 \rightarrow 29 \times 4$$

width with decimal point removed

length

$$30 \times 4$$

30 easier to work with than 29

$$1 \times 4$$

1 × 4 to be subtracted from 30 × 4

$$\begin{array}{r} 120 \\ - 4 \\ \hline 116 \end{array}$$

sum of 30×4

sum of 1×4

result of $120 - 4$

decimal point moved one place to the left to give answer 11.6

$$116 \rightarrow 11.6$$

First, take away the decimal point from the 2.9 to make the calculation 29×4 .

Change 29×4 to 30×4 since it is easier to work out. Write 1×4 below (the difference between 29×4 and 30×4).

Subtract 4 (product of 1×4) from 120 (product of 30×4) to give the answer of 116 (product of 29×4).

Move the decimal point one place to the left (it was moved one place to the right in the first step).

Top tricks

The multiplication tables of several numbers reveal patterns of multiplications. Here are two good mental tricks to remember when multiplying the 9 and 11 times tables.

multipliers 1 to 10

9 TIMES TABLE										
1	2	3	4	5	6	7	8	9	10	
9	18	27	36	45	54	63	72	81	90	

1 + 8 = 9 7 + 2 = 9 multiples of 9

△ Two digits are added together

The two digits that make up the first 10 multiples of 9 each add up to 9. The first digit of the multiple (such as 1, in 18) is always 1 less than the multiplier (2).

multipliers 1 to 9

11 TIMES TABLE									
1	2	3	4	5	6	7	8	9	
11	22	33	44	55	66	77	88	99	

11 × 3 = 33, or 3 written twice 11 × 7 = 77, or 7 written twice multiples of 11

△ Digit is written twice

To multiply by 11, merely repeat the two multipliers together. For example, 4×11 is two 4s or 44. It works all the way up to $9 \times 11 = 99$, which is 9 written twice.